

SARNA, Sweden

WMO: 023160

Lat: 61.691N

Lon: 13.187E

Elev: 1394

StdP: 13.97

Time Zone: 1.00 (EUC)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.9	(c) -17.9	(d) -28.4	(e) 1.2	(f) -23.8	(g) -22.3	(h) 1.7	(i) -17.8	(j) 27.5	(k) 33.0	(l) 23.3	(m) 21.7	(n) 1.1	(o) 320	(p) 0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	75.6	59.6	71.9	57.1	68.5	55.7	61.1	71.5	59.1	68.7	57.2	65.6	5.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.1	73.2	63.3	54.8	67.4	62.1	52.8	62.4	60.5	27.8	71.4	26.3	69.1	25.1	65.8	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.4	(b) 16.3	(c) 15.1	(e) -30.4	(f) 80.2	(g) 7.6	(h) 3.1	(i) -35.9	(j) 82.5	(k) -40.3	(l) 84.3	(m) -44.6	(n) 86.1	(o) -50.2	(p) 88.3		
(a) 22.4	(b) 16.3	(c) 15.1	(e) -30.6	(f) 83.6	(g) 7.5	(h) 2.5	(i) -36.0	(j) 85.4	(k) -40.4	(l) 86.9	(m) -44.6	(n) 88.3	(o) -50.1	(p) 90.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.8	13.5	16.0	23.6	31.9	42.9	51.8	55.7	52.4	44.3	35.0	22.9	13.8
	DBStd	17.70	14.82	13.33	9.68	7.25	6.70	5.64	5.39	6.00	5.65	7.46	10.40	13.62
	HDD50	6366	1132	952	817	544	240	41	11	43	189	464	812	1121
	HDD65	11412	1597	1372	1282	994	686	397	293	392	622	929	1262	1586
	CDD50	435	0	0	0	0	19	95	187	117	17	0	0	0
	CDD65	6	0	0	0	0	0	1	4	1	0	0	0	0
	CDH74	159	0	0	0	0	6	49	76	28	0	0	0	0
	CDH80	8	0	0	0	0	0	5	3	0	0	0	0	0
	WSAvg	4.2	4.0	4.5	4.8	4.1	4.6	4.7	4.7	4.2	4.1	3.9	3.4	3.8
	PrecAvg	26.2	1.6	1.1	1.1	1.6	2.2	3.1	3.3	3.8	2.5	2.4	2.0	1.5
Precipitation	PrecMax	34.6	3.5	2.1	2.3	4.1	4.5	6.5	5.5	6.3	4.4	5.3	5.5	2.6
	PrecMin	21.5	0.3	0.2	0.2	0.2	0.3	0.2	0.3	1.9	0.9	0.6	0.7	0.2
	PrecStd	3.1	0.8	0.5	0.6	1.1	1.2	1.3	1.2	1.3	1.0	1.3	1.1	0.6
	PrecStd	3.1	0.8	0.5	0.6	1.1	1.2	1.3	1.2	1.3	1.0	1.3	1.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	43.8	47.3	59.8	74.8	79.6	80.0	77.2	66.5	54.5	44.4	39.6
		MCWB	36.5	37.9	40.6	47.4	56.2	63.3	62.3	60.1	56.3	48.4	40.6	36.5
	2%	DB	37.8	40.4	42.8	53.0	70.4	73.2	75.9	73.6	61.5	50.9	41.3	37.5
		MCWB	34.1	36.2	36.1	43.2	53.9	58.3	60.1	58.9	51.6	45.8	38.7	34.8
	5%	DB	34.8	37.3	39.5	47.8	64.7	69.0	71.8	68.8	57.8	48.7	38.4	35.0
		MCWB	32.0	33.8	34.1	39.5	51.0	55.3	57.7	56.8	50.0	44.3	36.3	32.7
	10%	DB	32.6	34.1	36.8	44.0	59.0	65.0	68.4	64.2	54.7	46.0	35.2	32.8
		MCWB	30.6	31.3	33.0	37.7	47.6	53.2	56.4	55.0	48.4	42.4	33.6	30.9
	0.4%	WB	37.0	38.9	41.6	48.4	57.7	63.7	63.8	62.3	57.2	50.0	41.1	37.3
		MCDB	40.2	42.4	46.2	58.6	71.4	78.1	75.2	72.5	63.2	52.7	43.2	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.6	36.5	37.1	43.8	54.7	59.9	61.4	60.1	53.7	47.2	38.8	35.3
		MCDB	37.3	39.9	40.6	51.7	68.2	70.2	72.2	69.8	58.4	49.7	41.0	37.0
	5%	WB	32.5	34.0	35.0	40.6	51.8	56.9	59.3	58.2	51.4	44.7	36.4	32.9
		MCDB	34.3	37.0	38.3	46.1	63.4	66.3	68.8	66.8	56.2	48.0	38.1	34.6
	10%	WB	30.7	31.7	33.1	38.5	48.6	54.3	57.6	56.0	49.0	42.8	33.8	31.0
		MCDB	32.5	33.8	36.2	43.3	58.3	62.9	66.1	63.0	53.8	45.5	35.0	32.8
Mean Daily Temperature Range	5% DB	MDBR	16.2	19.3	19.0	18.0	21.8	19.1	19.5	18.6	17.4	13.2	12.0	16.0
		MCDBR	16.1	14.7	19.1	25.4	35.0	29.2	27.5	26.5	22.7	17.1	12.7	17.6
		MCWBR	14.4	12.1	13.8	16.1	19.7	15.8	13.5	13.7	13.8	12.6	10.9	15.5
	5% WB	MCDBR	15.5	14.6	16.4	21.8	33.8	25.5	22.8	22.2	16.9	14.1	11.2	16.5
		MCWBR	14.4	12.7	12.3	14.3	19.3	14.5	12.1	12.0	10.7	11.3	10.0	14.9
Clear-Sky Solar Irradiance	taub		0.212	0.263	0.291	0.327	0.337	0.332	0.348	0.339	0.308	0.292	0.230	0.341
	taud		2.124	2.266	2.272	2.339	2.458	2.499	2.464	2.481	2.551	2.491	2.212	2.824
	Ebn at Noon		166	230	262	274	281	284	276	269	259	221	157	118
	Edh at Noon		11	19	28	32	31	30	30	28	22	16	10	7
All-Sky Solar Radiation	RadAvg		87	293	787	1308	1492	1641	1492	1158	751	335	104	46
	RadStd		10	56	75	124	137	140	193	104	90	56	25	5

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page